

Estimation with TMLE in R

From Navigator plan to targeted estimate

Module G · Estimation

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The Navigator Carries the Study Plan into Estimation

Scientific question

In the synthetic pneumonia cohort, what is the effect of baseline pneumococcal vaccination on 12-month hospitalization risk?

Target

$$\Psi = E\left[Y^{(1)}\right] - E\left[Y^{(0)}\right]$$

A marginal risk difference.

Observed data

- A : baseline vaccination
- Y : hospitalization by 12 months
- W : age, prior pneumonia, prior vaccine

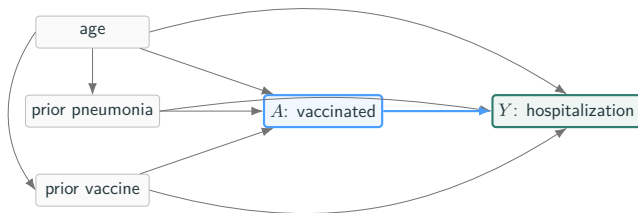
Estimation holds the population, outcome, adjustment set, and time horizon constant.

The Pneumonia Example Uses a Defined Cohort and Adjustment Set

Fixed synthetic cohort

Participants	5,000
Vaccinated	1,627
Unvaccinated	3,373
Hospitalizations	334

The age distribution is centered near 65. The sufficient adjustment set is {age, prior pneumonia, prior vaccine}.



The same baseline causes help explain vaccination and pneumonia risk.

The Crude Association Reflects Confounding by Indication

7.19%

vaccinated risk

—

6.43%

unvaccinated risk

Crude RD

+0.76 percentage points

Participants with higher baseline pneumonia risk were more likely to be vaccinated. The crude contrast therefore mixes the treatment effect with baseline differences between treatment groups.

Four Estimators Address the Same Risk-Difference Target

Estimator	Main model	How it forms the contrast
Crude	None	Compares the two observed treatment groups
G-computation	$Q(A, W) = E(Y \mid A, W)$	Predicts outcomes under $A = 1$ and $A = 0$, then averages
Hajek IPW	$g(W) = P(A = 1 \mid W)$	Reweights observed outcomes to balance measured W
TMLE	Q and g	Targets the outcome fit for Ψ , then uses the influence curve

The comparison is informative because the target stays fixed

Only the estimation strategy changes.

G-Computation Standardizes Outcome Predictions

Learn the conditional outcome mean:

$$Q(A, W) = E(Y \mid A, W).$$

For every participant, obtain

$$\hat{Q}(1, W_i) \quad \text{and} \quad \hat{Q}(0, W_i).$$

Then standardize over the cohort:

$$\hat{\Psi}_{\text{gcomp}} = \frac{1}{n} \sum_{i=1}^n \left\{ \hat{Q}(1, W_i) - \hat{Q}(0, W_i) \right\}.$$

In this example

Parametric g-computation:

−3.28 pp

Standardization compares vaccination strategies in the same measured covariate distribution.

The Treatment Model Estimates Vaccination Probability

Learn the treatment mechanism:

$$g(W) = P(A = 1 \mid W).$$

It describes how vaccination varies with measured baseline history.

Hajek IPW uses $g(W)$ to reweight observed outcomes:

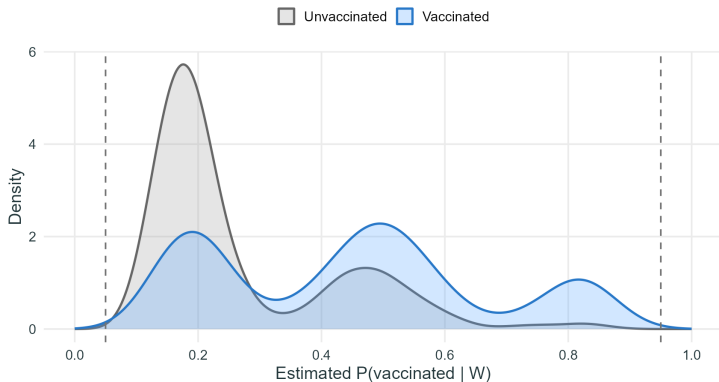
$$\hat{\Psi}_{\text{IPW}} = -2.94 \text{ pp.}$$

TMLE also uses $g(W)$ through the clever covariate:

$$H(A, W) = \frac{A}{g(W)} - \frac{1 - A}{1 - g(W)}.$$

$H(A, W)$ connects treatment assignment to the residuals from the initial outcome model.

Overlap Informs the Positivity Assessment



Observed diagnostic

Estimated propensity range:

0.077 to 0.909.

No fitted values fall outside
[0.05, 0.95].

The treatment groups share support across the measured covariate distribution in this synthetic cohort.

TMLE Combines the Outcome and Treatment Models



The targeting step makes a focused update to the initial outcome predictions for the chosen risk-difference parameter.

Double robustness uses information from both nuisance models; causal interpretation still relies on the Navigator's identification assumptions.

A Compact R Implementation Produces the Targeted Estimate

```
d <- read.csv("data/pneumonia_data.csv")
W <- d[c("age", "priorPneumonia", "priorVaccine")]
A <- d$A
Y <- d$Y

learners <- c("SL.glm", "SL.glmnet", "SL.ranger")
set.seed(2026)
fit <- tmle(
  Y = Y, A = A, W = W,
  Q.SL.library = learners,
  g.SL.library = learners,
  V.Q = 3, V.g = 3
)

fit$estimates$ATE$psi
fit$estimates$ATE$CI
```

Full live script: pneumonia_tmle_example.R.

Diagnostics and Reproducibility Accompany the Estimate

Data and target

- Fixed 5,000-row synthetic CSV
- A , Y , and baseline W coded explicitly
- Marginal 12-month risk difference

Support

- Propensity range: 0.077 to 0.909
- Treatment-specific densities overlap

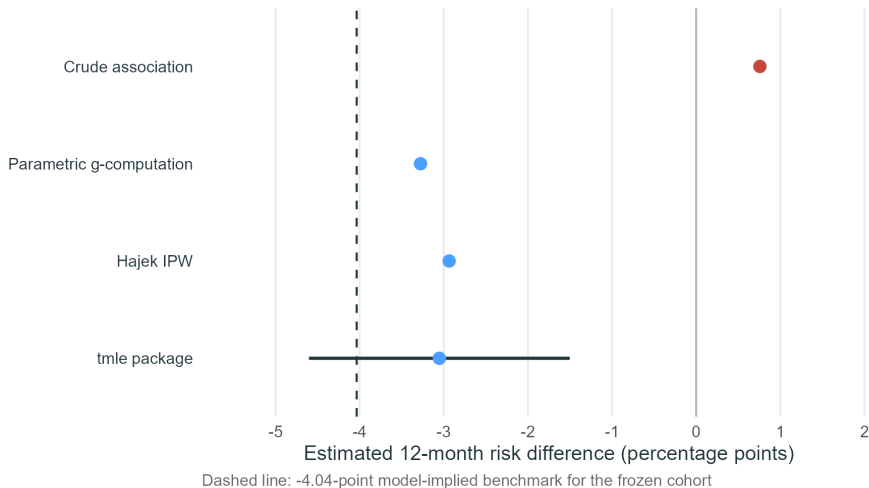
Nuisance learning

- GLM, penalized GLM, and random forest
- Three-fold cross-validation for workshop runtime

Audit trail

- Standalone generator and value check
- Saved results, diagnostics, figures, and session info

Adjustment Changes the Estimated Direction and Magnitude



The adjusted estimators are similar in this synthetic example. That agreement supports an implementation check; it does not assess unmeasured confounding.

The TMLE Estimate Is a 3.05-Percentage-Point Risk Reduction

—3.05 pp

95% CI: —4.60 to —1.50 pp

Interpretation

If the stated assumptions identify the target, vaccination is estimated to reduce 12-month pneumonia hospitalization risk by about 3.05 events per 100 people in this synthetic cohort.

The model-implied benchmark for the frozen cohort is —4.04 percentage points. It is available here only because the teaching data generator is known.

Synthetic teaching result - not clinical evidence.

TMLE Inherits the Study's Causal Assumptions

Identification

- Conditional exchangeability given measured W
- Positivity in the target population
- Consistency of the treatment and outcome definitions

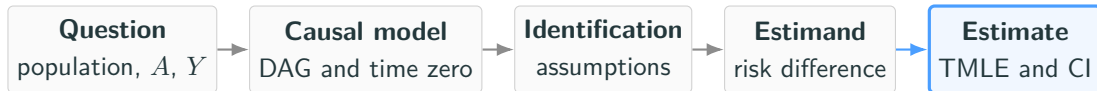
Study implementation

- Accurate treatment and outcome measurement
- A common time zero and 12-month follow-up
- Appropriate handling of missingness and selection

Scope

These assumptions belong to the study and causal model. TMLE carries them into an estimator with flexible nuisance fitting and influence-curve uncertainty.

The Estimate Completes the Roadmap



One continuous analysis

The same study plan now connects the scientific question to a reproducible estimate, uncertainty interval, and assumption-calibrated interpretation.

Next resource: the technical supplement opens the Q , g , targeting, and influence-curve steps.